

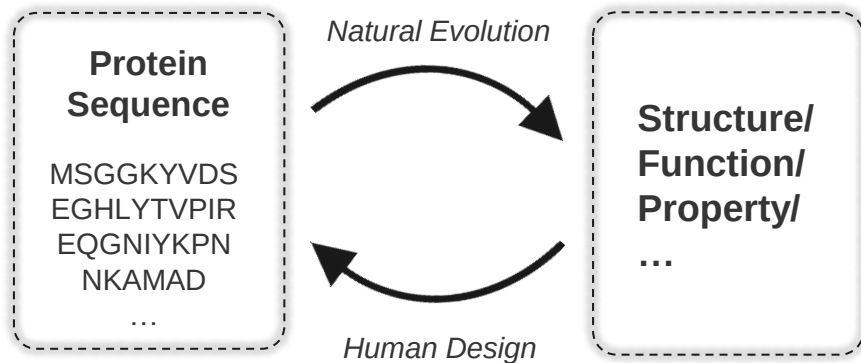
PDAgent: An LLM-Driven Autonomous Agent Framework Towards *In Silico* Protein Design via Directed Mutation

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What is Protein Design?



Challenges:

- **VAST** Search Space (20^{100} potential sequences for a 100-AAs protein)
- Gap between **Structure/Function/Property** and **Sequence**
- Dynamic and Flexibility
- and so on...



Biomedical

- De Novo Antibodies
- Vaccine Design
- Immune Engineering
- ...



Industry

- Biocatalysis
- Biomaterials
- Biosensors
- ...

Protein Design



Research

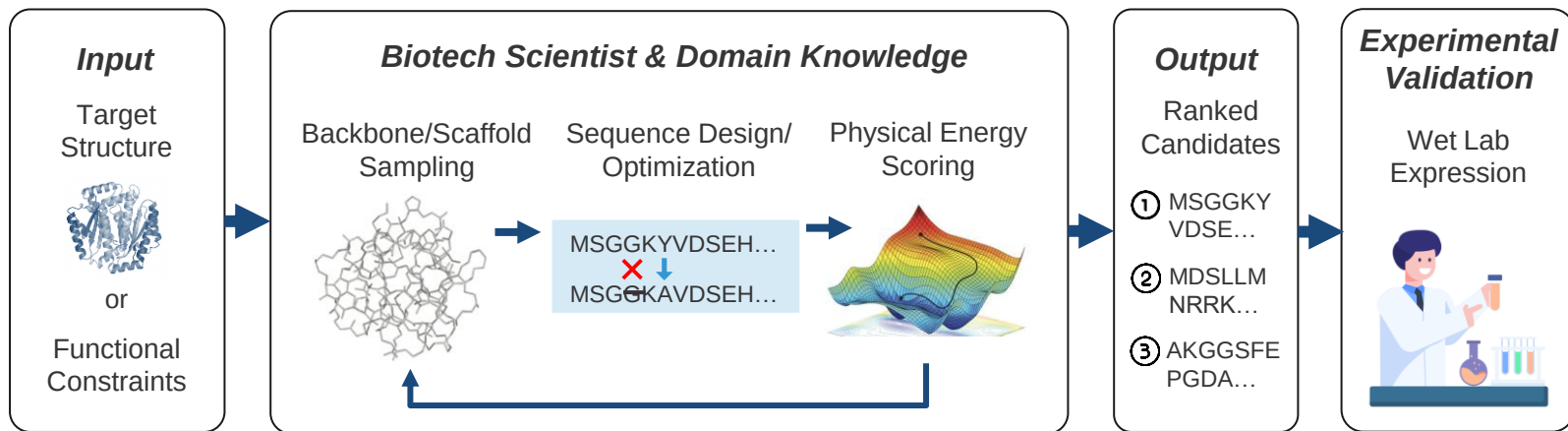
- Interaction Prediction
- Fitness Landscape
- De Novo Folding
- ...



Environment

- Bioremediation
- Carbon Sequestration
- Plastic-Degrading Enzymes
- ...

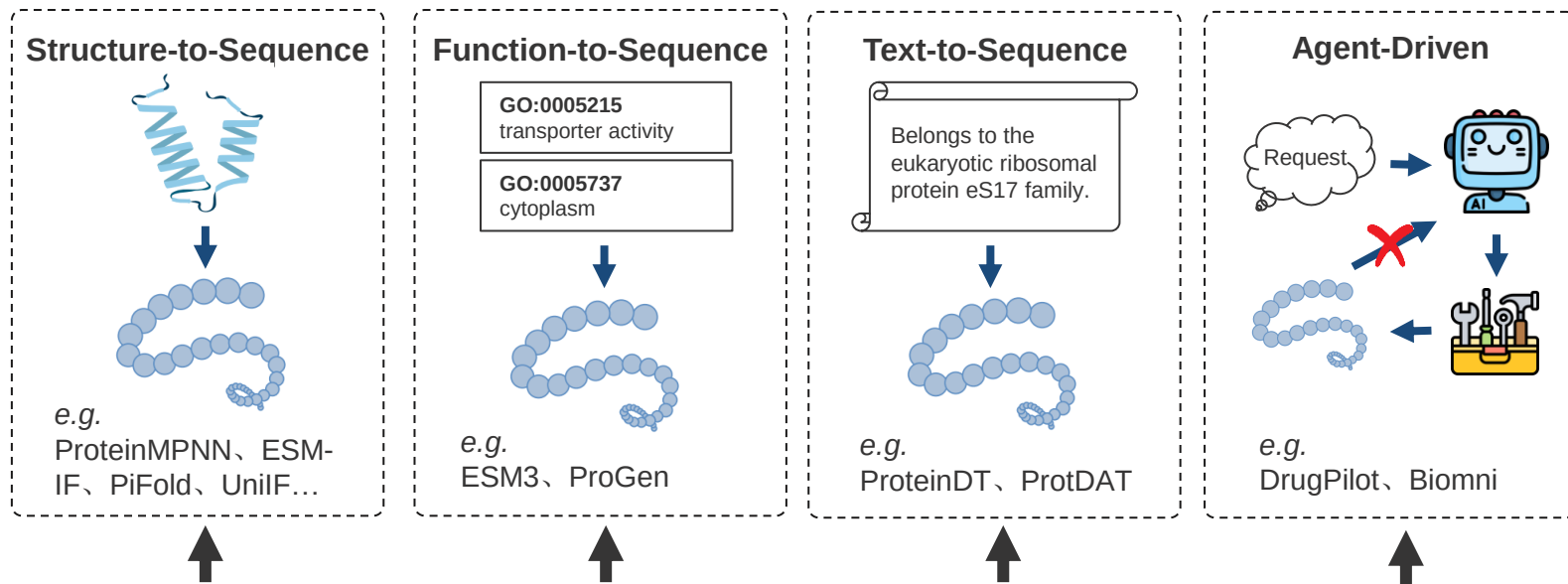
How do traditional approaches operate?



Limitations:

- **High barrier:** Requires multidisciplinary background in structural biology, bioinformatics, and computational chemistry
- **Fragmented toolchains:** Various software tools operate independently
- **Low efficiency:** Manual operations are time-consuming and limited ability to explore mutation space

How about AI-powered approaches?



Limitations:

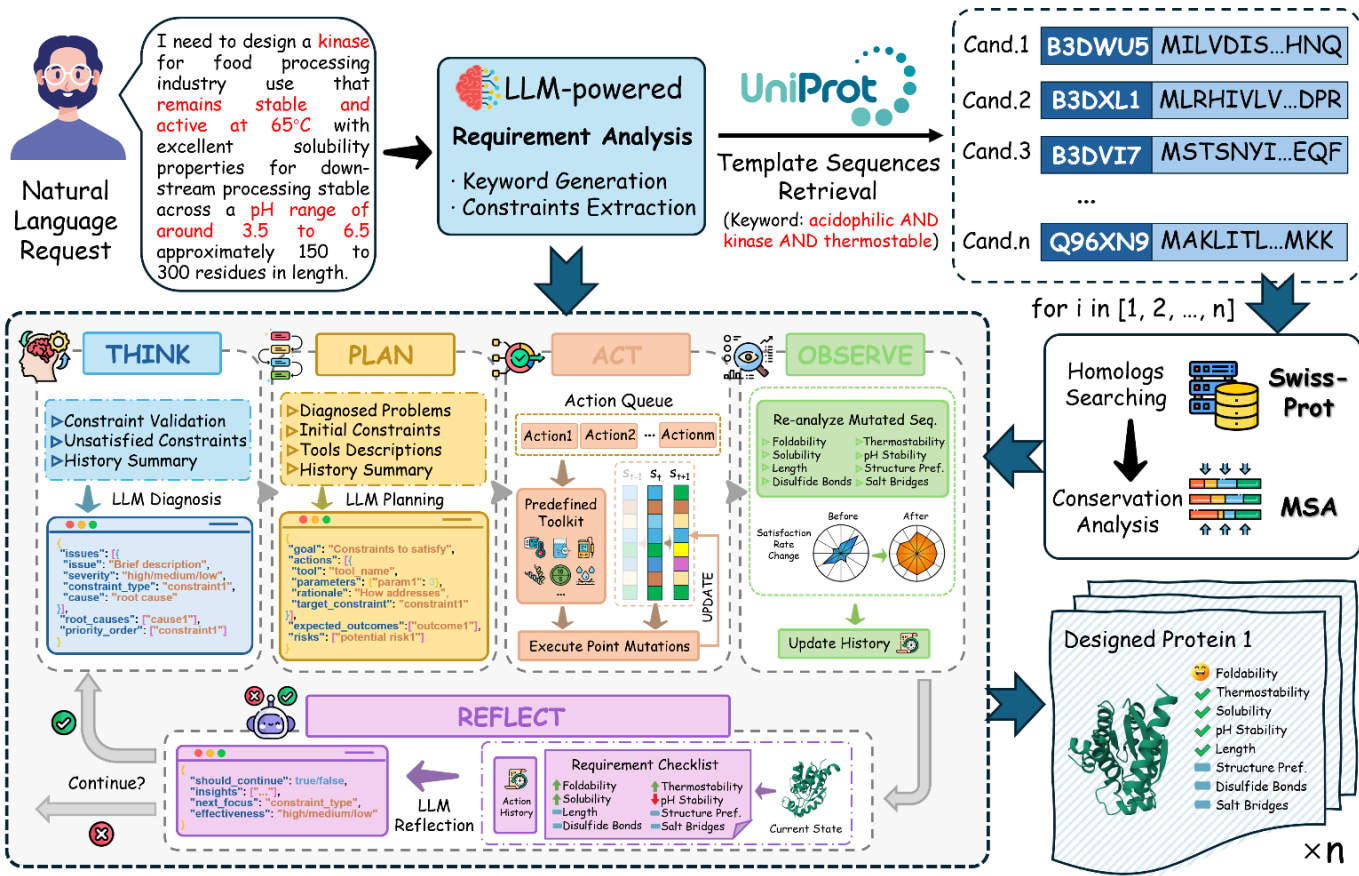
Lacking explicit functional regularization

Relying on predefined functional ontologies

Intrinsic gap between language and sequence

Lacking closed-loop feedback/self-optimization

PDAgent Framework

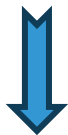


PDAgent - Unified Constraint Representation

Natural
Language
Request



I need to design a **kinase** for food processing industry use that **remains stable and active at 65° C** with excellent solubility properties for down-stream processing stable across a **pH range of around 3.5 to 6.5** approximately 150 to 300 residues in length.



- Constraints Extraction
- Keyword Generation

Constraints:

Type: thermostability
Target_value: 65
Comparison: >=

Type: length
Target_value: null
Comparison: range
Range_min: 150
Range_max: 300

Type: solubility
Target_value: True
Comparison: boolean

Type: length
Target_value: null
Comparison: range
Range_min: 150
Range_max: 300

Keywords:

- kinase
- thermostable
- acidophilic

Constraint Type List

- ◆ Thermostability
- ◆ Solubility
- ◆ pH Stability
- ◆ Length
- ◆ Structure Preference
- ◆ Disulfide Bonds
- ◆ Salt Bridges

PDAgent - Candidate Sequences Preparation

- Candidates Retrieval

Keywords:

- kinase
- thermostable
- acidophilic



✓ **Candidate 1**

B3DWU5:MILVDISNSVTKIGLFKEGELRLLEKIP...

✓ **Candidate 2**

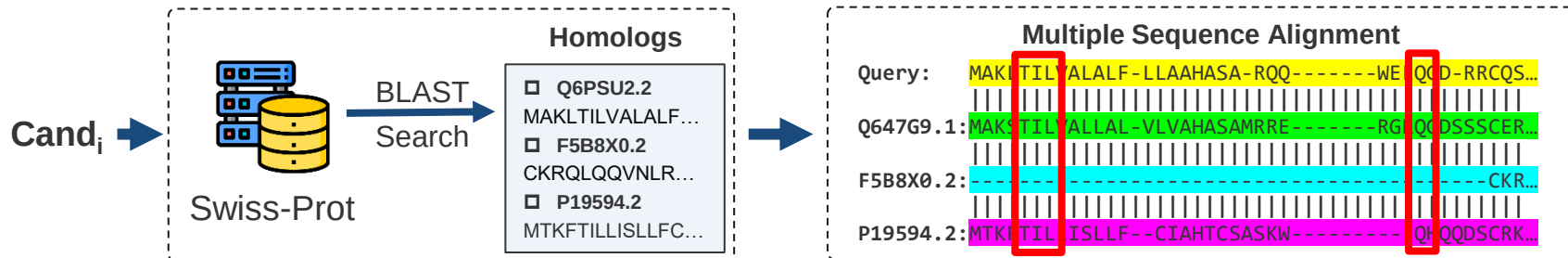
B3DXL1:MLRHIVLVGMMGAGKTTVGLWLSKAKSI...

✓ **Candidate 3**

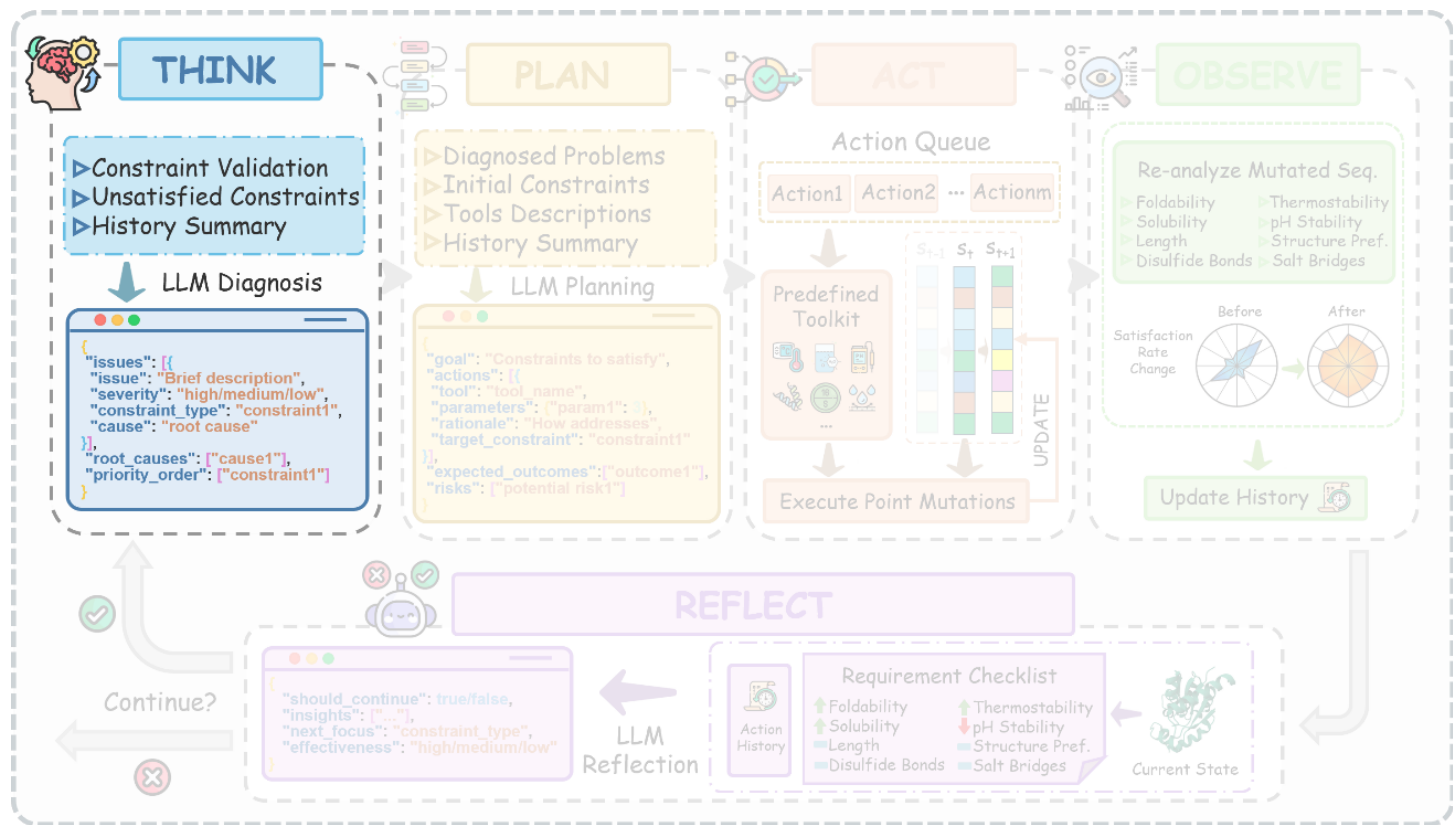
B3DVI7:MSTSNYINSFAPAKINLGLRIFGKRKDG...

... ..

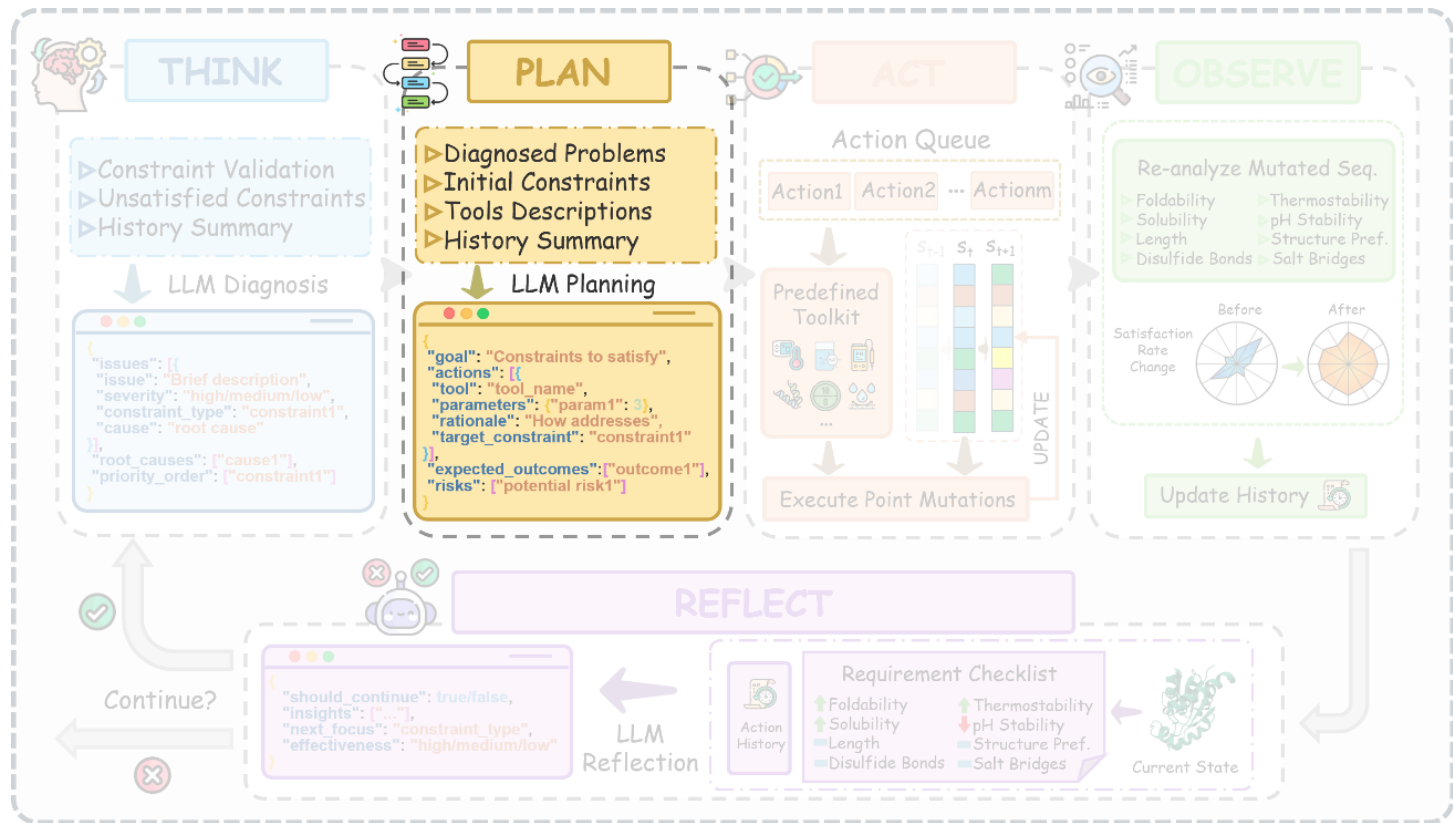
- Conservation Analysis



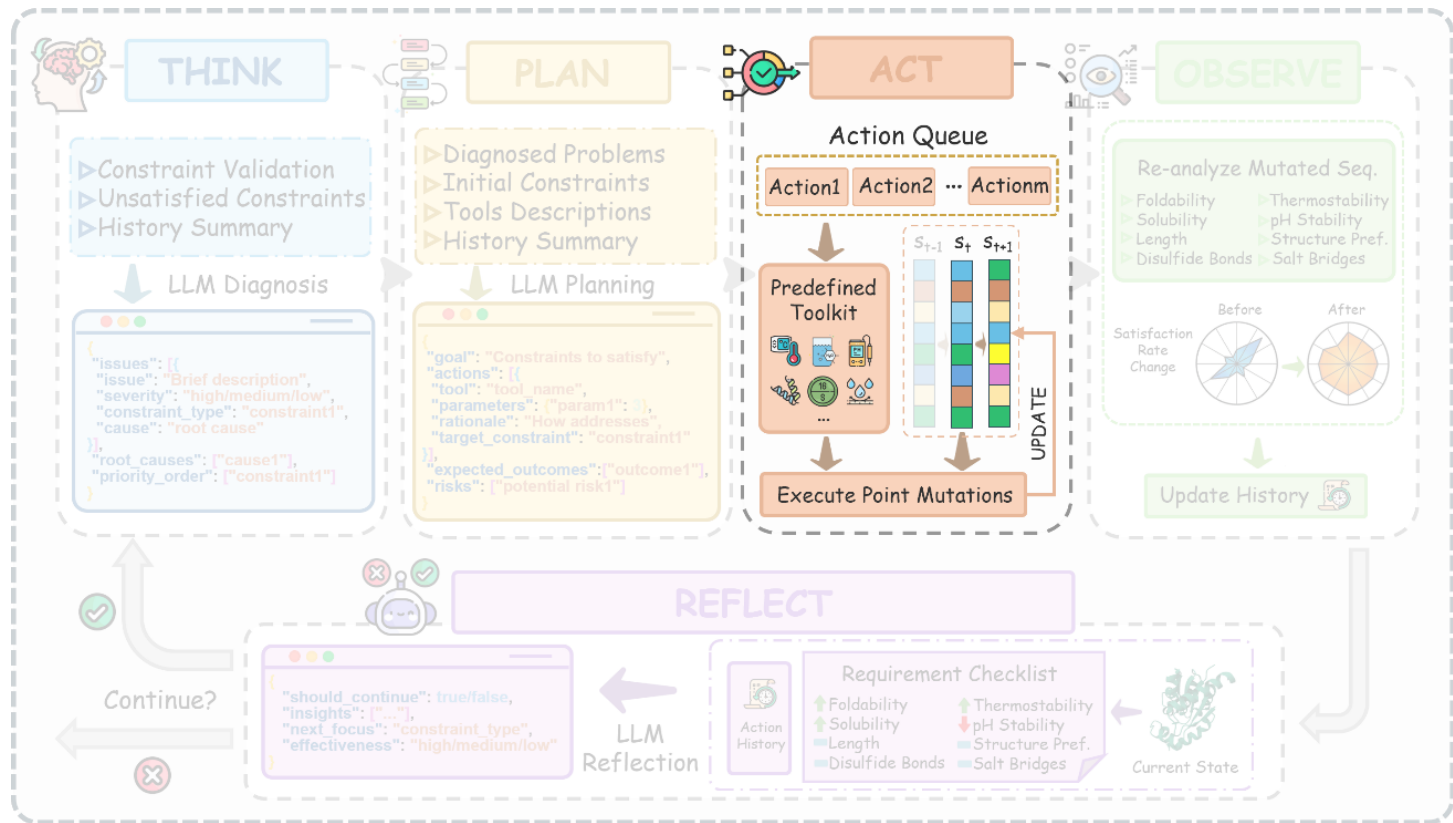
PDAgent - THINK



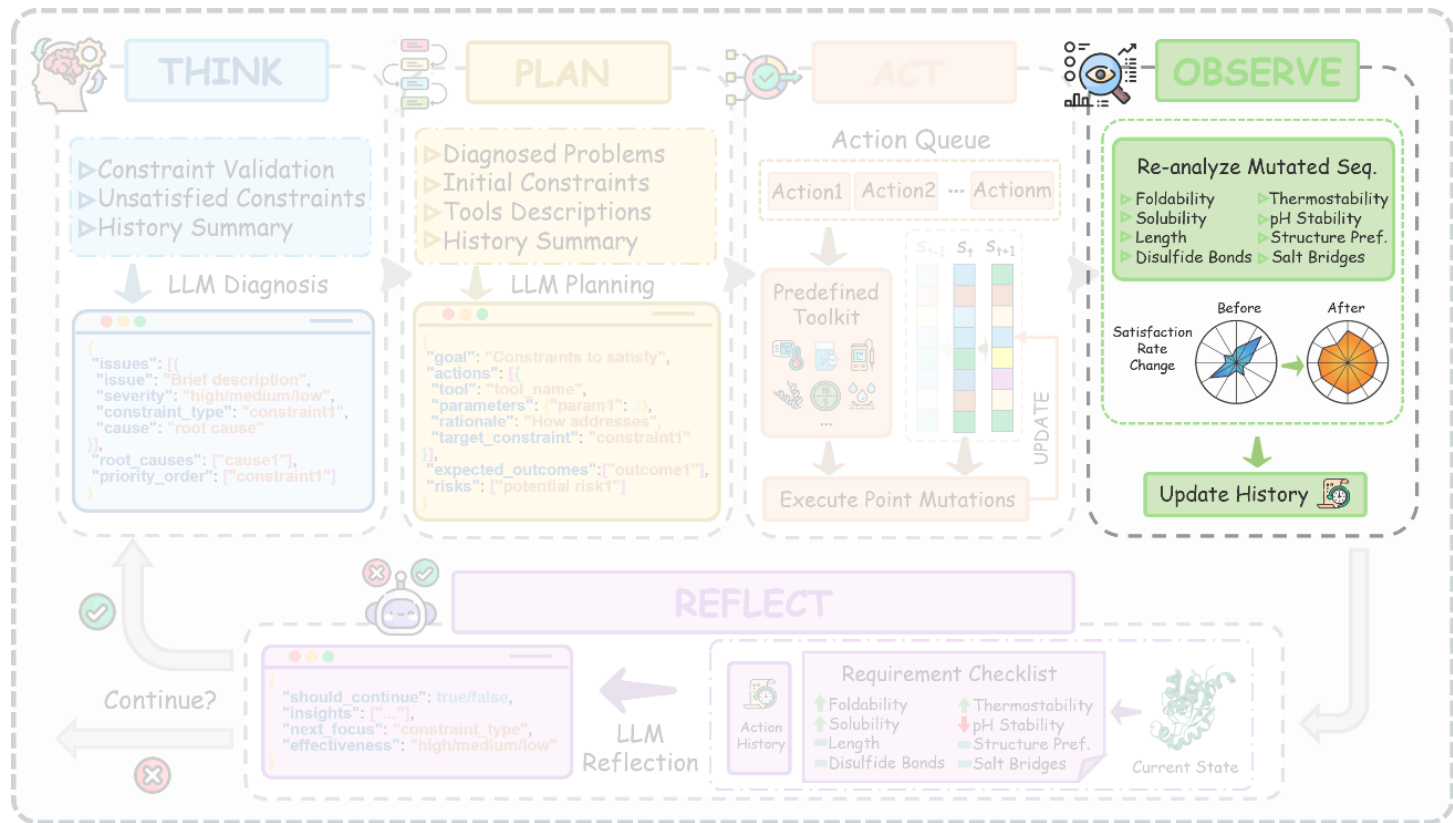
PDAgent - PLAN



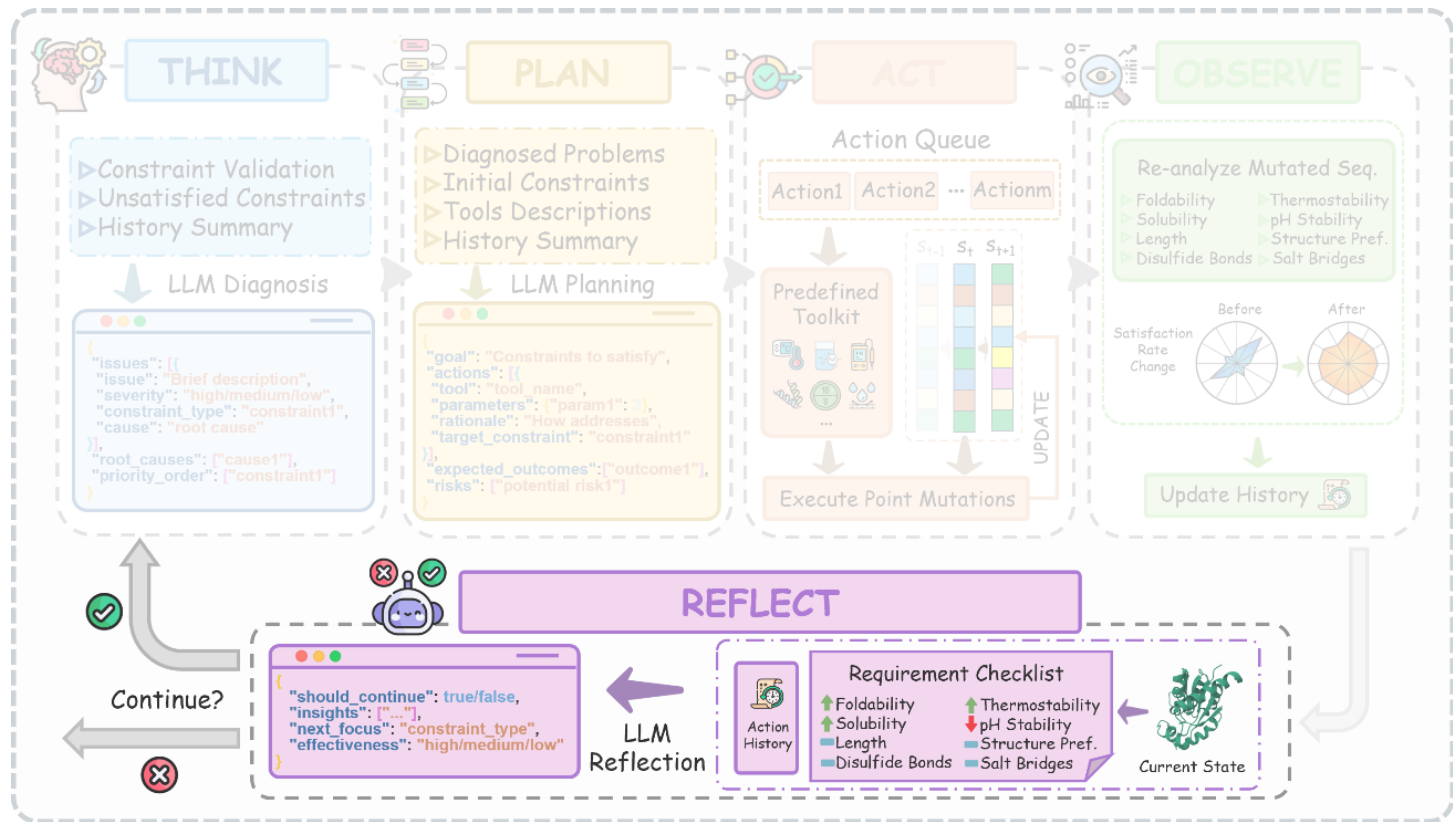
PDAgent - ACT



PDAgent - OBSERVE



PDAgent - REFLECT



Experiments

- Main Results

Model	SR (%)	Fold.	CSR (%)							
			Thermo.	Solu.	pH	Length	Struct.	Disulf.	Salt	Avg.
<i>LLM Directly Prompting Methods</i>										
GPT-4o	96	37.62	29.87	78.46	77.78	81.48	81.25	90.00	100.00	76.98
GPT-5.2	93	47.84	28.77	82.81	73.68	76.92	<u>96.67</u>	87.50	100.00	78.05
o4-mini	62	27.50	22.45	43.18	63.64	100.00	90.48	100.00	100.00	74.25
Gemini-3-Pro	95	50.03	60.53	61.54	58.82	100.00	<u>96.67</u>	100.00	100.00	82.55
Claude-Sonnet-4.5	76	30.52	26.23	100.00	<u>88.24</u>	30.00	82.14	100.00	100.00	75.23
Claude-Haiku-4.5	98	42.99	16.67	18.18	42.11	10.34	81.82	63.64	100.00	47.54
DeepSeek-V3.2	100	52.86	20.00	72.05	36.84	3.45	51.52	0.00	77.78	37.38
Grok-4	99	45.59	37.97	89.55	47.37	51.72	84.38	54.55	100.00	66.51
<i>Other Protein Design Methods</i>										
ProtDAT	100	46.91	3.75	66.18	53.33	3.45	39.39	27.27	100.00	41.91
ProteinDT	100	39.15	2.50	66.18	40.00	10.34	60.61	27.27	88.89	42.26
Pinal	100	<u>80.92</u>	14.29	64.71	53.33	13.79	63.64	63.64	100.00	53.34
BioMedGPT-10B	85	<u>72.88</u>	25.76	100.00	0.00	26.92	48.15	0.00	0.00	28.69
Biomni	50	35.08	5.26	88.57	41.67	40.00	55.56	57.14	100.00	55.46
<i>Ours</i>										
PDAgent (LLaMA-3.1-8B)	100	87.69	<u>46.25</u>	100.00	<u>93.99</u>	100.00	93.94	100.00	100.00	<u>90.60</u>
PDAgent (Qwen-3-30B)	100	80.54	<u>55.00</u>	100.00	80.00	100.00	87.88	72.73	100.00	<u>85.09</u>
PDAgent (DeepSeek-V3.2)	100	87.69	46.15	100.00	100.00	100.00	96.88	100.00	100.00	91.86



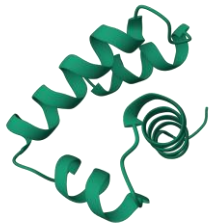
Experiments

- Case Study



I need to engineer a lipase enzyme for diagnostic assay development with high thermostability to **withstand 65 °C** with enhanced solubility in aqueous solutions engineered for **alkaline environments at pH 8.5** rich in **alpha-helical content**.

GPT-4o



Baseline
Gallery

Length: 66
☹️ Mean pLDDT: 52.6
✗ Tm: 57.585°C
✗ GRAVY: 0.0833
✅ Isoelectric Point: 9.3007
✅ Helix/Sheet/Turn:
56.06%/39.39%/10.61%

GPT-5.2



Length: 124
😊 Mean pLDDT: 91.95
✗ Tm: 51.529°C
✅ GRAVY: -0.4137
✗ Isoelectric Point: 4.7715
✅ Helix/Sheet/Turn:
100.00%/25.81%/0.00%

o4-mini

Failed

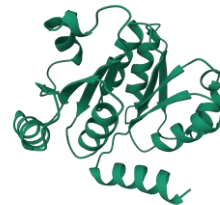
Length: -
☹️ Mean pLDDT: -
✗ Tm: -
✗ GRAVY: -
✗ Isoelectric Point: -
✗ Helix/Sheet/Turn:
-%/-/%-

Gemini 3 Pro



Length: 92
☹️ Mean pLDDT: 60.85
✅ Tm: 67.862°C
✗ GRAVY: 0.0348
✅ Isoelectric Point: 7.0096
✅ Helix/Sheet/Turn:
66.30%/38.04%/11.96%

Claude
Sonnet 4.5



Length: 203
☹️ Mean pLDDT: 53.03
✗ Tm: 55.486°C
✅ GRAVY: -0.2084
✗ Isoelectric Point: 5.8255
✅ Helix/Sheet/Turn:
40.89%/29.56%/25.62%

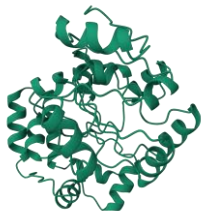
Experiments

- Case Study



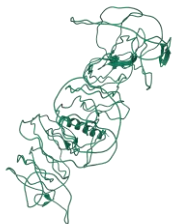
I need to engineer a lipase enzyme for diagnostic assay development with high thermostability to **withstand 65 °C** with enhanced solubility in aqueous solutions engineered for **alkaline environments at pH 8.5** rich in **alpha-helical content**.

Claude
Haiku 4.5



Length: 288
☹️ Mean pLDDT: 33.69
✗ Tm: 59.673°C
✗ GRAVY: 0.4340
✗ Isoelectric Point: 9.4847
✅ Helix/Sheet/Turn:
50.00%/38.5%/20.14%

DeepSeek-
V3.2



Length: 500
☹️ Mean pLDDT: 22.27
✗ Tm: 54.165°C
✗ GRAVY: 0.0833
✅ Isoelectric Point: 10.0001
✗ Helix/Sheet/Turn:
2.60%/2.40%/96.40%

Grok-4



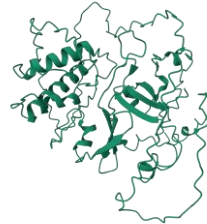
Length: 165
☹️ Mean pLDDT: 47.06
✅ Tm: 76.570°C
✅ GRAVY: -0.4897
✗ Isoelectric Point: 5.3959
✅ Helix/Sheet/Turn:
64.24%/29.09%/12.12%

ProtDAT



Length: 500
☹️ Mean pLDDT: 49.95
✗ Tm: 49.277°C
✗ GRAVY: 0.1888
✗ Isoelectric Point: 4.0500
✗ Helix/Sheet/Turn:
5.40%/25.00%/41.00%

ProteinDT



Length: 426
☹️ Mean pLDDT: 32.48
✗ Tm: 46.908°C
✅ GRAVY: -0.0892
✗ Isoelectric Point: 5.8046
✗ Helix/Sheet/Turn:
28.87%/38.73%/39.67%

Baseline
Gallery

Experiments

- Case Study



I need to engineer a lipase enzyme for diagnostic assay development with high thermostability to **withstand 65 °C** with enhanced solubility in aqueous solutions engineered for **alkaline environments at pH 8.5** rich in **alpha-helical content**.

Pinal



😊 Length: 414
Mean pLDDT: 83.09
❌ Tm: 39.950°C
✅ GRAVY: -0.2879
❌ Isoelectric Point: 5.6826
❌ Helix/Sheet/Turn:
31.16%/36.23%/27.05%

BioMedGPT-10B



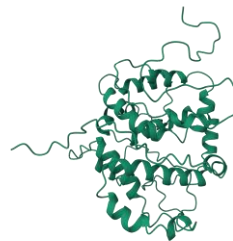
😞 Length: 239
Mean pLDDT: 73.0402
❌ Tm: 56.541°C
❌ GRAVY: 0.0833
❌ Isoelectric Point: 11.0000
✅ Helix/Sheet/Turn:
100.00%/0.00%/0.00%

Biomni

Failed

😞 Length: -
Mean pLDDT: -
❌ Tm: -
❌ GRAVY: -
❌ Isoelectric Point: -
❌ Helix/Sheet/Turn:
-%/-/%

Random



😞 Length: 326
Mean pLDDT: 25.75
❌ Tm: 42.731°C
✅ GRAVY: -0.6856
✅ Isoelectric Point: 8.4975
❌ Helix/Sheet/Turn:
24.54%/32.52%/24.54%

Baseline
Gallery

Experiments

- Case Study



I need to engineer a lipase enzyme for diagnostic assay development with high thermostability to **withstand 65 ° C** with enhanced solubility in aqueous solutions engineered for **alkaline environments at pH 8.5** rich in **alpha-helical content**.

Recognized Constraints

Type: thermostability
Target_value: 65
Comparison: >=

Type: solubility
Target_value: True
Comparison: boolean

Type: pH Stability
Target_value: 8.5
Comparison: ==

Type: structure pref.
Target_value: alpha
Comparison: ==

UniProt
Searching
&
Conservation
Analysis



UniProt ID: Q56221

Sequence:

MG**F**DDKQDFLEETFAK**Y**PEGRRAI
MPL**L**RRVQEE**G**WIRPERIEEIARLVGT
TPTE**V**MG**V**ASFYSYYQFV**P**TGKYHLQV
CATLS**C**KLGAEEELWDYLTET**L**GIGPGE
VTP**D**GL**F**SVQKVE**C**LG**S**CHT**A**PVIQVN
DEPY**V**ECVTRARLEALLAG**L**RAG**K**RLE
EIL**P**KG**C**GHH**V**HE**V**

Mean pLDDT: 87.25



Tm: 77.592°C



GRAVY: -0.1911



Isoelectric Point: 4.9992



Helix/Sheet/Turn:



34.81%/37.57%/21.55%

adjust_pi:

D5K, D6K

stabilize_helix:

G54L, P108E, P20A, P140L

enhance_solubility:

A93S, L54E



Bold Red: Highly conserved positions
(based on 29 homologs)

Experiments

• Case Study

Target:

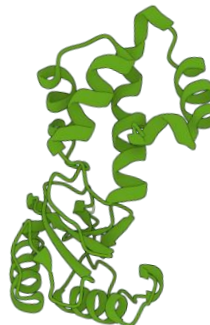
Type: thermostability Target_value: 65 Comparison: >=	Type: solubility Target_value: True Comparison: boolean	Type: pH stability Target_value: 8.5 Comparison: ==	Type: structure pref. Target_value: alpha Comparison: ==
--	--	--	---



adjust_pl:
Q8K, D9K
enhance_solubility:
A25S, L140R, V73T



adjust_pl:
E21K, S25K
enhance_solubility:
L168E, A84S
introduce_disulfide:
T55C, T73C



adjust_pl:
Q36K, E37K
enhance_solubility:
L163E, F10Y, F72Y



Sequence:

MGFFKKKKQDLEETFAKYPAGEGRRRAAIM
PLLRRVQEEGWIRPERIEEIALRLVETTP
TEVMGVASFYSYYQFVPTGKYHLQVCAT
LSCKLAGSEELWDYLTETLGIGEGEVTP
DGLFSVQKVECLGSCHTAPVIQVNDERY
VECVTRARLEALLAGLRAGKRLEEIEELPG
KCGHHVHEVEV

Mean pLDDT: 84.3

😊 Tm: 75.317°C

✓ GRAVY: -0.1890

✓ Isoelectric Point: 5.2200

✗ Helix/Sheet/Turn:

✗ 37.57%/38.12%/18.79%

Sequence:

MGFFKKKKQDLEETFAKYPAGEGRRSAIM
PLLRRVQEEGWIRPERIEEIALRLVETTP
TEVMGVASFYSYYQFTPTGKYHLQVCAT
LSCKLAGSEELWDYLTETLGIGEGEVTP
DGLFSVQKVECLGSCHTAPVIQVNDERY
VECVTRARLEALLAGLRAGKRLEEIEELPG
KCGHHVHEVEV

Mean pLDDT: 85.40

😊 Tm: 76.310°C

✓ GRAVY: -0.2807

✓ Isoelectric Point: 5.8058

✗ Helix/Sheet/Turn:

✓ 37.58%/37.57%/18.78%

Sequence:

MGFFKKKKQDLEETFAKYPAGRRKAIM
PLLRRVQEEGWIRPERIEEIALRLVETTP
TEVMGVASFYSYYQFCPTGKYHLQVCST
LSCKLAGSEELWDYLTETLGIGEGEVTP
DGLFSVQKVECLGSCHTAPVIQVNDERY
VECVTRARLEALLAGLRAGKRLEEIEEPG
KCGHHVHEVEV

Mean pLDDT: 82.56

😊 Tm: 76.835°C

✓ GRAVY: -0.3193

✓ Isoelectric Point: 6.2431

✗ Helix/Sheet/Turn:

✓ 37.57%/35.91%/18.78%

Sequence:

MGFFKKKKQDLEETFAKYPAGRRKAIM
PLLRRVQKEGWIRPERIEEIALRLVETTP
TEVMGVASFYSYYQYCPGKYHLQVCS
TLSCCKLAGSEELWDYLTETLGIGEGEVTP
DGLFSVQKVECLGSCHTAPVIQVNDERY
VECVTRARLEALLAGLRAGKRLEEIEEP
GKCGHHVHEVEV

Mean pLDDT: 88.15

😊 Tm: 76.822°C

✓ GRAVY: -0.4094

✓ Isoelectric Point: 6.9502

✓ Helix/Sheet/Turn:

✓ 38.12%/35.35%/18.78%

Conclusion

- ✓ We propose PDAgent, an LLM-driven autonomous agent framework for natural language-guided protein design that enables iterative, closed-loop sequence optimization through a five-phase ReAct cycle.
- ✓ We integrate conservation analysis with stability-aware mutation filtering to ensure generated sequences maintain structural integrity while satisfying user-specified design constraints.
- ✓ We demonstrate through experiments on 100 diverse design tasks that PDAgent achieves 91.86% average constraint satisfaction rate with high mean pLDDT, substantially outperforming both direct LLM generation and specialized design methods.

Thanks for watching!



Project

Also feel free to contact me at ouyangsong@whu.edu.cn